

Last time : $a > 0$

$$\cdot \int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin\left(\frac{u}{a}\right) + C$$

where $|u| < a$

$$\cdot \int \frac{du}{a^2 + u^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

$$\cdot \int \frac{du}{u \sqrt{u^2 - a^2}} = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$$

$|u| > a$

$$\underline{Ex} : \int \frac{dx}{\sqrt{5 - x^2 + 4x}} = ?$$

Completing the square :

$$\begin{aligned} 5 - x^2 + 4x &= 5 - (x-2)^2 + 4 \\ &= 9 - (x-2)^2 \\ &= 3^2 - (x-2)^2 \end{aligned}$$

$$u = x - 2$$

$$\int \frac{dx}{\sqrt{5 - x^2 + 4x}} = \int \frac{du}{\sqrt{3^2 - u^2}}$$

$$= \arcsin\left(\frac{u}{3}\right)$$

+ C

$$= \arcsin\left(\frac{x-2}{3}\right) + C$$

$$\underline{Ex} : \quad I = \int_0^{\pi/4} \sqrt{1 + \cos(4x)} \, dx$$

Note : $1 + \cos(4x) = 2 \cos^2(2x)$

$$I = \sqrt{2} \int_0^{\pi/4} \underbrace{|\cos(2x)|}_{\geq 0} \, dx$$

$$= \sqrt{2} \int_0^{\pi/4} \cos(2x) \, dx$$

$$= \sqrt{2} \left[\frac{\sin(2x)}{2} \right]_0^{\pi/4}$$

$$= \frac{\sqrt{2}}{2}$$

Ex : Reducing a fraction

$$\int \frac{3x^2 - 2x + 1}{3x + 1} \, dx = ?$$

P, Q polynomials
 $Q \neq 0$

Then $\exists A, B$ polynomials
s.t. $\deg(B) < \deg(Q)$

and $P = AQ + B$ (Euclidean
division
of P by Q)
 $B = \text{remainder}$

$$3x^2 - 2x + 1 = A \cdot (3x + 1) + \underbrace{\text{constant}}_B$$

$$3x^2 - 2x + 1 - x(3x + 1) = -3x + 1$$

$$-3x + 1 - (-1)(3x + 1) = 2$$

Conclusion:

$$3x^2 - 2x + 1 = (x - 1)(3x + 1) + 2$$

$$\int \frac{3x^2 - 2x + 1}{3x + 1} dx = \int (x - 1) dx + 2 \int \frac{1}{3x + 1} dx$$

$$= \frac{(x - 1)^2}{2} + \frac{2}{3} \log |3x + 1|$$

Integration by part

Technique to compute $\int f(x)g(x)dx$

Let F be an antiderivative of f .

Note: $(F \cdot g)' = f \cdot g + F \cdot g'$

So $f \cdot g = (F \cdot g)' - F \cdot g'$

Therefore:

$$\int f(x)g(x)dx = F(x)g(x) - \int F(x)g'(x)dx$$

Integration by part method.

Ex : $\int x \sin(x) dx = ?$

$$f(x) = \sin(x)$$

$$g(x) = x$$

$$F(x) = -\cos(x)$$

$$g'(x) = 1$$

$$\begin{aligned}\int x \sin(x) dx &= -\cos(x) \cdot x \\ &\quad - \int (-\cos(x)) \cdot 1 dx \\ &= -\cos(x) \cdot x + \int \cos(x) dx \\ &= -\cos(x) \cdot x + \sin(x) + C\end{aligned}$$



if we had chosen

$$f(x) = x \quad \text{and} \quad g(x) = \sin(x)$$

We would get

$$f(x) = \frac{x^2}{2}$$

$$g'(x) = \cos(x)$$

$$\int x \sin(x) dx = \frac{x^2}{2} \cdot \sin(x) - \int \frac{x^2}{2} \cos(x) dx$$

∴
More complicated

⇒ Need to choose f and g
"carefully" when trying an
integration by part

f should be as "simple" as
possible

g'

Sometimes, we may need to repeat the integration by part several times.

Ex: $\int x^2 e^x dx = ?$

$$f(x) = e^x$$

$$g(x) = x^2$$

$$F(x) = e^{2x}$$

$$g'(x) = 2x$$

$$\text{So } \int x^2 e^x dx = e^x \cdot x^2 - \int e^x \cdot (2x) dx$$

$$\int e^x \cdot x dx = ?$$

Again integration by part

$$f(x) = e^x$$

$$g(x) = x$$

$$F(x) = e^x$$

$$g'(x) = 1$$

$$\begin{aligned}\int e^x \cdot x \, dx &= e^x \cdot x - \int e^x \cdot 1 \, dx \\ &= e^x \cdot x - e^x\end{aligned}$$

$$\text{So } \int x^2 e^x \, dx = e^x \cdot x^2 - 2e^x \cdot x + 2e^x$$

The same technique would work
for $\int x^n e^x \, dx$, $\forall n \in \mathbb{N}$
(n integration by part).

$$\underline{\text{Ex}}: \int e^x \cos(x) \, dx = ?$$

$$\underline{\text{Try}}: \begin{array}{ll} f(x) = e^x & F(x) = e^x \\ g(x) = \cos(x) & g'(x) = -\sin(x) \end{array}$$

$$\int e^x \cos(x) \, dx = e^x \cos(x) - \int e^x \cdot (-\sin(x)) \, dx$$

$$\int e^x \sin(x) dx = e^x \sin(x) - \int e^x \cos(x) dx$$

So by combining the two, we get

$$\int e^x \cos(x) dx = e^x \cos(x) + \left(e^x \sin(x) - \int e^x \cos(x) dx \right)$$

So

$$\int e^x \cos(x) dx = \frac{e^x \cos(x) + e^x \sin(x)}{2}$$

Ex : $n \geq 1$ $I_n = \int (\cos(x))^n dx$

$$I_n = \int f(x) g(x) dx \text{ where } f(x) = \cos(x) \\ g(x) = (\cos(x))^{n-1}$$

$$F(x) = \sin(x)$$

$$g'(x) = (n-1) (-\sin(x)) \cos(x)^{n-2}$$

$$\text{So } I_n = \sin(x) \cdot \cos(x)^{n-1} \\ - \int \sin(x) \cdot (n-1) (-\sin(x)) \cdot \cos(x)^{n-2} dx$$

$$\sin(x)^2 = 1 - \cos(x)^2$$

$$\text{So } \sin(x) \cdot (n-1) \cdot (-\sin(x)) \cdot \cos(x)^{n-2} \\ = -(n-1) \cdot (\cos(x)^{n-2} - \cos(x)^n)$$

$$I_n = \sin(x) \cos(x)^{n-1} + (n-1) \cdot I_{n-2} \\ - (n-1) I_n$$

$$\text{So } n \cdot I_n = \sin(x) \cos(x)^{n-1} \\ + (n-1) \cdot I_{n-2}$$

By induction, we can compute I_{n-2} .

Sometimes, the choice of f and g may not be obvious.

Ex: $\int \log(x) dx = ?$

$$\log(x) = f(x) \cdot g(x)$$

$$f(x) = 1$$

$$g(x) = \log(x)$$

$$F(x) = x \quad g'(x) = \frac{1}{x}$$

$$\int \log(x) dx = x \cdot \log(x) - \int x \cdot \frac{1}{x} dx$$

$$\int \log(x) dx = x \log(x) - x + C$$

Rational functions and partial fractions

$$\int f(x) dx \quad \text{where } f(x) = \frac{P(x)}{Q(x)}$$

where $P(x)$, $Q(x)$ are
polynomials.

$$\underline{\text{Ex}}: \int \frac{3x-4}{(x-1)(x-2)} dx = ?$$

$$\underline{\text{Trick}}: \frac{3x-4}{(x-1)(x-2)} = \frac{1}{x-1} + \frac{2}{x-2}$$

$$\text{So: } \int \frac{3x-4}{(x-1)(x-2)} = \int \frac{1}{x-1} dx + 2 \int \frac{1}{x-2} dx$$

$$= \log|x-1| + 2\log|x-2|$$

More generally; given a rational function $f(x) = \frac{P(x)}{Q(x)}$, one may

write it as a sum of "partial fractions"

Simplest situation :

$$Q(x) = \prod_{i=1}^n (x - \alpha_i) \quad \begin{array}{l} \alpha_i \in \mathbb{R} \\ \text{distincts} \end{array}$$

product

$$= (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n)$$

Without loss of generality,

one can assume that $\deg P < \deg Q$
(otherwise, divide P by Q)

Prop : $\exists A_1, \dots, A_n \in \mathbb{R}$
(unique) s.t

$$f(x) = \sum_{i=1}^n \frac{A_i}{x - \alpha_i} \quad (*)$$

We don't prove it.

Question : How do you compute
the A_i 's ?

Fix $i \in \{1, \dots, n\}$.

Multiply both sides of (*)
by $(x - \alpha_i)$. Then

evaluate both sides at $x = \alpha_i$

We get :

$$A_i = \frac{P(\alpha_i)}{\prod_{\substack{j=1 \\ j \neq i}}^n (\alpha_i - \alpha_j)}$$

Ex : $\frac{x^2 + 4x + 1}{(x-1)(x+1)(x+3)}$

$$= \frac{A_1}{x-1} + \frac{A_2}{x+1} + \frac{A_3}{x+3}$$

$A_1 = ?$ Multiply both sides
by $x-1$

$$A_1 = \frac{1^2 + 4 \cdot 1 + 1}{(1+1)(1+3)} = \frac{6}{2 \cdot 4} = \frac{3}{4}$$

$$A_2 = \frac{1}{2}$$

$$A_3 = -\frac{1}{4}$$

Conclusion :

$$\frac{x^2 + 4x + 1}{(x-1)(x+1)(x+3)} = \frac{3}{4} \cdot \frac{1}{x-1} + \frac{1}{2} \cdot \frac{1}{x+1} - \frac{1}{4} \cdot \frac{1}{x+3}$$

$$\text{So } \int \frac{x^2 + 4x + 1}{(x-1)(x+1)(x+3)} dx$$

$$= \frac{3}{4} \log|x-1| + \frac{1}{2} \log|x+1| - \frac{1}{4} \log|x+3|.$$

—————→

Trigonometric integrals

$$\underline{\text{Ex}}: \int f(x) dx$$

where $f(x) = \text{rational function}$
in $\sin(x)$ and

$$\text{i.e. } f(x) = \frac{\sum_{i=1}^n \lambda_i \cos(x)^{a_i} \sin(x)^{b_i}}{\sum_{j=1}^m \mu_j \cos(x)^{c_j} \sin(x)^{d_j}}$$

e.g. $\int \frac{1}{\cos(x)^2} dx = \int (\sec(x))^2 dx$
 $= \tan(x) + c$

$$\underline{\text{Ex}}: \int \sin(x)^m \cos(x)^n dx$$

where $m, n \in \mathbb{Z}_{\geq 0}$

(Recall: we did $m=0$ using integration by part)

① If m or n is odd

Then e.g. $m = 2k + 1$

$$\begin{aligned}\sin(x)^m &= \sin(x)^{2k+1} \\ &= \sin(x)^{2k} \cdot \sin(x) \\ &= (1 - \cos(x)^2)^k \cdot \sin(x)\end{aligned}$$

$$\begin{aligned}I &= \int \sin(x)^m \cos(x)^n dx \\ &= - \int (1 - \cos(x)^2)^k \cos(x)^n d(\cos(x))\end{aligned}$$

$$u = \cos(x)$$

$$I = - \int \underbrace{(1 - u^2)^k u^n}_{\text{polynomial in } u \leadsto \text{easy to compute}} du$$

② If both m and n are even

$$m = 2a$$

$$n = 2b$$

$$\sin(x)^m = \sin(x)^{2a} = \left(\frac{1 - \cos(2x)}{2} \right)^a$$

$$\cos(x)^n = \left(\frac{1 + \cos(2x)}{2} \right)^b$$

Expand $\sin(x)^m \cos(x)^n$

$\Rightarrow$ get lower powers of $\cos(2x)$

$$\underline{\text{Ex}}: \int \sin(x)^4 \cos(x)^2 dx$$

$$= \int \left(\frac{1 - \cos(2x)}{2} \right)^2 \left(\frac{1 + \cos(2x)}{2} \right) dx$$

$= \dots$

$$= \frac{1}{4} \int \frac{1 - \cos(2x)^2 + \cos(2x)^3 - \cos(2x)}{2} dx$$

$$u = 2x$$

$$= \frac{1}{8} \int \frac{1 - \cos(u) - \cos(u)^2 + \cos(u)^3}{2} du$$

$$\int \cos(u) du \quad \text{and} \quad \int \cos(u)^3 du$$

are easy

(odd exponent)

$$\int \cos(u)^2 du = \int \frac{1 + \cos(2u)}{2} du$$

= ...

You may also use the
linearization technique:

Express $\cos(x)^n \sin(x)^m$

in terms of linear combinations
of $\cos(ax)$ and $\sin(bx)$

where $a, b \in \mathbb{Z}$

Relies on:

$$\begin{cases} \sin(\alpha) \cos(\beta) = \frac{\sin(\alpha+\beta) + \sin(\alpha-\beta)}{2} \\ \sin(\alpha) \sin(\beta) = \frac{\cos(\alpha-\beta) - \cos(\alpha+\beta)}{2} \\ \cos(\alpha) \cos(\beta) = \frac{\cos(\alpha+\beta) + \cos(\alpha-\beta)}{2} \end{cases}$$

Ex: linearize $\cos(x)^3$

$$\cos(x)^3 = \frac{1}{4} \cos(3x) + \frac{3}{4} \cos(x)$$

$$\cos(x)^2 \cos(x)$$

$$\cos(x)^2 = \frac{1 + \cos(2x)}{2}$$

$$\cos(x)^2 \cdot \cos(x) = \frac{(1 + \cos(2x)) \cos(x)}{2}$$

linearize

$$\cos(2x) - \cos(x)$$

$$= \frac{\cos(3x) + \cos(x)}{2}$$

Trigonometric substitutions

These are useful substitutions
when we deal with

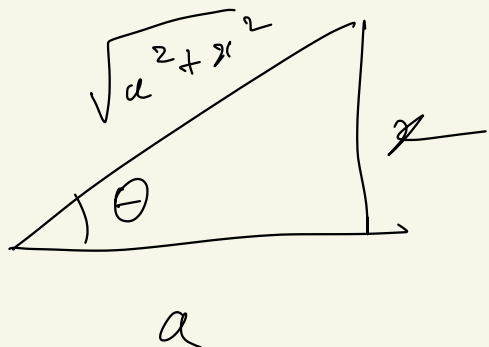
expressions of the form

$$\sqrt{a^2 - x^2} \quad \text{or} \quad \sqrt{a^2 + x^2}$$

$$\text{or} \quad \sqrt{x^2 - a^2}$$

$$(a > 0) \quad (x = \text{variable})$$

$$\sqrt{a^2 + x^2}$$



$$\tan(\theta) = \frac{x}{a}$$

$$x = a \tan(\theta)$$

$$\sqrt{a^2 + x^2} = a \sec(\theta)$$

We want the change of variable to be reversible

We restrict $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$$dx = a(\sec(\theta))^2 d\theta$$

$$\underline{E_x} : \int \frac{dx}{\sqrt{4 + x^2}}$$

$$a = 2$$

$$x = 2 \tan(\theta)$$

$$\sqrt{4+x^2} = 2 \sec \theta$$

$$dx = 2 (\sec(\theta))^2 d\theta$$

$$\int \frac{dx}{\sqrt{4+x^2}} = \int \frac{2 (\sec \theta)^2 d\theta}{2 \sec(\theta)}$$

$$= \int \sec(\theta) d\theta$$

$$= ??$$

Formula: $\int \sec(\theta) d\theta$

$$= \log | \sec(\theta) + \tan(\theta) |$$

(check) $+ C$

We get:
$$\int \frac{dx}{\sqrt{4+x^2}} = \log \left| \frac{\sec(\theta)}{1+\tan(\theta)} \right| + C$$

$$\int \frac{dx}{\sqrt{4+x^2}} = \log \left| \frac{x}{2} + \sqrt{\frac{x^2}{4} + 1} \right| + C$$

$$\int \sec(x) dx = ?$$

$$= \int \frac{1}{\cos(x)} dx$$

$$\boxed{u = \sin(x)}$$

$$du = \cos(x) dx$$

Bioche's
rule

$$\int \frac{1}{\cos(x)} dx = \int \frac{du}{\cos(x)^2}$$

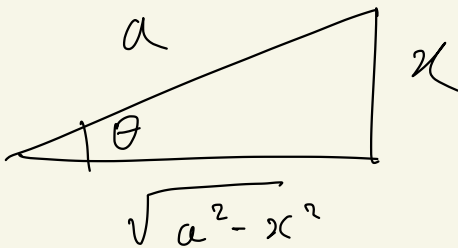
$$= \int \frac{du}{1-u^2}$$

$$= \frac{1}{2} \log \left| \frac{1+u}{1-u} \right|$$

$$= \frac{1}{2} \log \left| \frac{1+\sin(x)}{1-\sin(x)} \right|$$

$$= \log |\sec(x) + \tan(x)| + C$$

$$\sqrt{a^2 - x^2}$$



$$\sin(\theta) = \frac{x}{a}$$

$$\boxed{x = a \sin(\theta)}$$

$$\sqrt{a^2 - x^2} = a \cos(\theta) \quad dx = a \cos(\theta) d\theta$$

Restrict: $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

E_x:
$$I = \int \left(\sqrt{1-x^2} \right)^3 dx$$

$$x = \sin(\theta) \quad dx = \cos(\theta) d\theta$$

$$\sqrt{1-x^2} = \cos(\theta)$$

$$I = \int (\cos(\theta))^3 \cos(\theta) d\theta$$

$$= \int (\cos(\theta))^4 d\theta$$

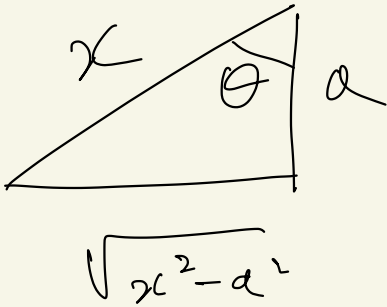
$\approx \dots$

We have
seen
that

(either linearization
or integration
by part)

$$\sqrt{x^2 - a^2}$$

$$|x| > a > 0$$



$$\cos(\theta) = \frac{a}{x}$$

$$x = a \cdot \sec(\theta)$$

We restrict for
simplicity to
 $\theta \in [0, \frac{\pi}{2})$

$$\sqrt{x^2 - a^2}$$

$$= a \cdot \tan(\theta)$$

$$\int \frac{dx}{\sqrt{25x^2 - 4}} \quad x > \frac{2}{5}$$

$$I = \quad = ?$$

$$\frac{1}{5} \int \frac{dx}{\sqrt{x^2 - \left(\frac{2}{5}\right)^2}}$$

$$a = \frac{2}{5}$$

$$x = \frac{2}{5} \sec(\theta)$$

$$dx = \frac{2}{5} \sec(\theta) \tan(\theta) d\theta$$

$$\sqrt{x^2 - \left(\frac{2}{5}\right)^2} = \frac{2}{5} \tan(\theta)$$

$$I = \frac{1}{5} \int \frac{\frac{2}{5} \sec(\theta) \tan(\theta) d\theta}{\frac{2}{5} \tan(\theta)}$$

$$= \frac{1}{5} \int \sec(\theta) d\theta$$

$$= \frac{1}{5} \log |\sec(\theta) + \tan(\theta)|$$

$$x = \frac{2}{5} \sec(\theta) \quad + \quad C$$

$$= \frac{1}{5} \log \left| \frac{5x}{2} + \sqrt{\frac{25}{4}x^2 - 1} \right|$$

$$+ C.$$